

Sports Betting Arbitrage Detection and Analysis System

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1 Motivation and Rationale

For this project, I built a data pipeline that collects sportsbook odds and checks whether the differences between books create arbitrage opportunities. In this context, an arbitrage opportunity means that the odds are different enough across sportsbooks that a bettor could theoretically cover both sides of a game and still make a guaranteed profit.

I chose this project because sportsbook odds change constantly, and different platforms do not always update at the same time. That makes the problem a good fit for web scraping, API collection, data cleaning, and automated analysis. The main challenge was not just collecting the data, but getting odds from different sources into one consistent format so they could be compared reliably.

The project also connected directly to several topics from the course, including APIs, scraping, data cleaning, integration, visualization, and analysis with Pandas.

2 Description of Data Sources

The project collected betting odds data from several sportsbooks using a combination of APIs and Selenium-based web scraping.

2.1 Data Sources

The following sportsbooks were used:

- DraftKings
- BetMGM
- FanDuel
- Bovada
- BetRivers

The project monitored odds for the following sports:

- NBA
- MLB
- NHL

2.2 Data Collection Methods

Two different collection methods were used:

1. **The Odds API** was used to retrieve structured sportsbook odds data for FanDuel, Bovada, and BetRivers.
2. **Selenium web scraping** was used for DraftKings and BetMGM because these websites rely heavily on dynamically generated JavaScript content.

I ran the collection process roughly every 90 seconds over several days. I focused on upcoming games instead of live games because live odds move too quickly and would make the arbitrage results less stable. For each sportsbook record, I kept the sportsbook name, sport, team name, decimal odds, event start time, and scrape timestamp.

- Team names
- Sportsbook name
- Decimal betting odds
- Event start time
- Sport category
- Timestamp of data collection

2.3 Challenges Encountered

Several challenges occurred during development:

- Different sportsbooks formatted team names differently.
- Some sportsbooks used accented characters such as “Montréal Canadiens.”
- Selenium scraping behaved differently in headless browser mode.
- Sportsbooks dynamically loaded odds data after page load.
- Odds APIs used different JSON structures than scraped websites.
- The biggest challenge involved maintaining Selenium scrapers over time.

The most difficult part of the project was keeping the Selenium scrapers working consistently. DraftKings and BetMGM both use dynamic pages, so the odds were not always available immediately when the page loaded. I had to use waits and scrolling logic to give the page enough time to load the betting markets before extracting the data.

Another issue was that the data did not look the same across sources. Team names were sometimes formatted differently, and some names included accented characters, such as “Montréal Canadiens.” To avoid mismatches, I added normalization functions that lowercased names, removed accents, and mapped team-name variations to a standard version.

The API data also had a different structure from the scraped data, so I used separate cleaning logic for API results and Selenium results before combining them into the same final format. The biggest limitation is that Selenium scrapers are fragile. A scraper that works one day can fail later if the website changes its HTML structure, class names, or loading behavior.

3 Integrated Data Model

After collection, I converted all sportsbook records into a common Pandas DataFrame. This was necessary because the API results and Selenium results originally used different structures. Once cleaned, every row represented one sportsbook’s odds for one team in one game. The primary data model contained the following attributes:

Field	Description
sportsbook	Name of sportsbook
sport	NBA, MLB, or NHL
game_id	Standardized event identifier
team	Team name
odds_decimal	Decimal betting odds
commence_time	Game start time (UTC)
status	Upcoming or live
scrape_time	Timestamp of collection

The key integration step was creating a shared game identifier. Since sportsbooks may list teams in different orders or use slightly different names, I normalized both team names, sorted them, and joined them into one event ID. For example, a Lakers vs. Warriors game became:

lakers__warriors

This made it possible to group odds from different sportsbooks under the same game and compare the best available odds for each side.

3.1 Entity-Relationship Diagram

Figure 1 shows the informal entity-relationship diagram used to model the relationships between sportsbooks, sporting events, teams, and arbitrage opportunities.

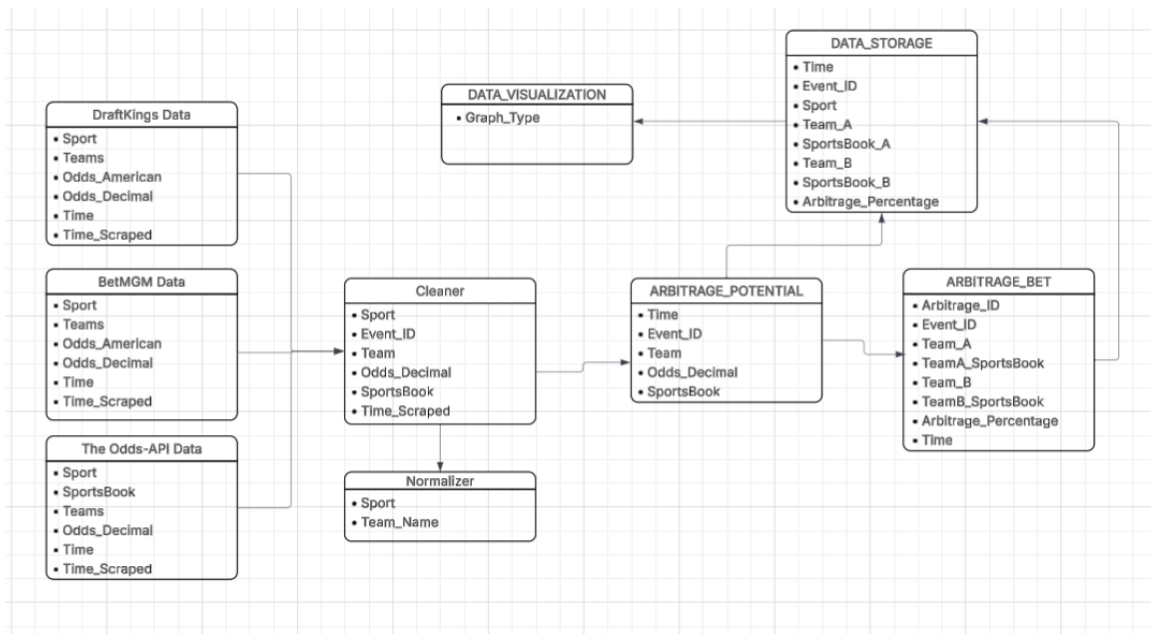


Figure 1: Informal Entity-Relationship Diagram

4 Analysis and Visualizations

Once the data was integrated, I grouped records by game and compared the available odds across sportsbooks. For each event, the system selected the best odds for each possible outcome. I then converted those odds into implied probabilities. For a two-team event, the implied probability calculation was:

$$\frac{1}{\text{Odds Team A}} + \frac{1}{\text{Odds Team B}}$$

If the total was below 1, then the difference represented an arbitrage opportunity. The profit margin was calculated as:

$$\text{Profit Margin} = \left(1 - \left(\frac{1}{\text{Odds Team A}} + \frac{1}{\text{Odds Team B}} \right) \right) \times 100$$

4.1 Frequency of Arbitrage Opportunities Over Time

Figure 1 shows how arbitrage opportunities fluctuated over time during data collection.

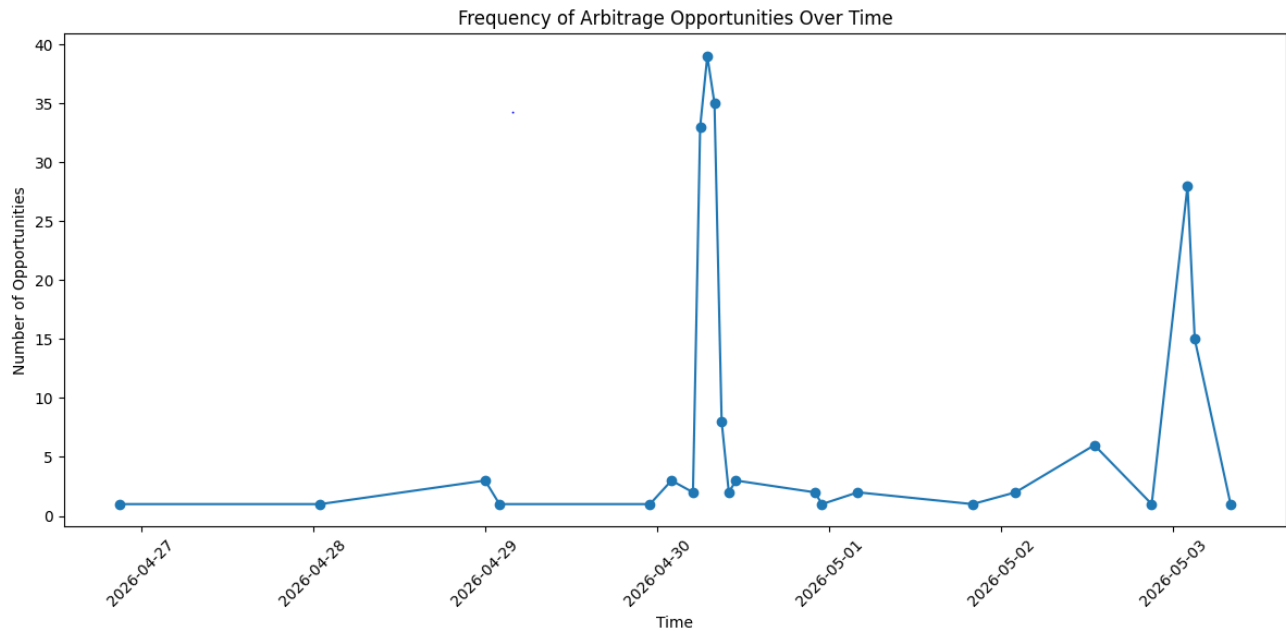


Figure 2: Frequency of Arbitrage Opportunities Over Time

The time-series graph shows that arbitrage opportunities did not appear evenly throughout the collection period. Instead, they appeared in short bursts, especially around certain dates. This suggests that the timing of data collection matters because many opportunities may only exist briefly before sportsbooks update their odds.

4.2 Arbitrage Opportunities by Sport

Figure 2 compares the number of arbitrage opportunities discovered across different sports.

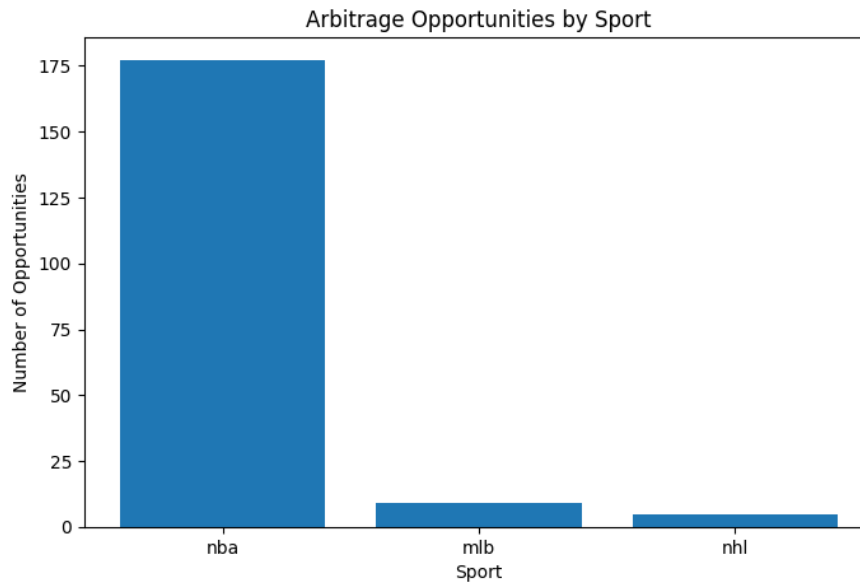


Figure 3: Arbitrage Opportunities by Sport

NBA produced the largest number of arbitrage opportunities in this dataset. One possible explanation is that NBA games receive more betting activity, which may cause sportsbooks to update lines more often and create short periods where their odds disagree.

4.3 Distribution of Profit Margins

Figure 3 displays the distribution of arbitrage profit margins.

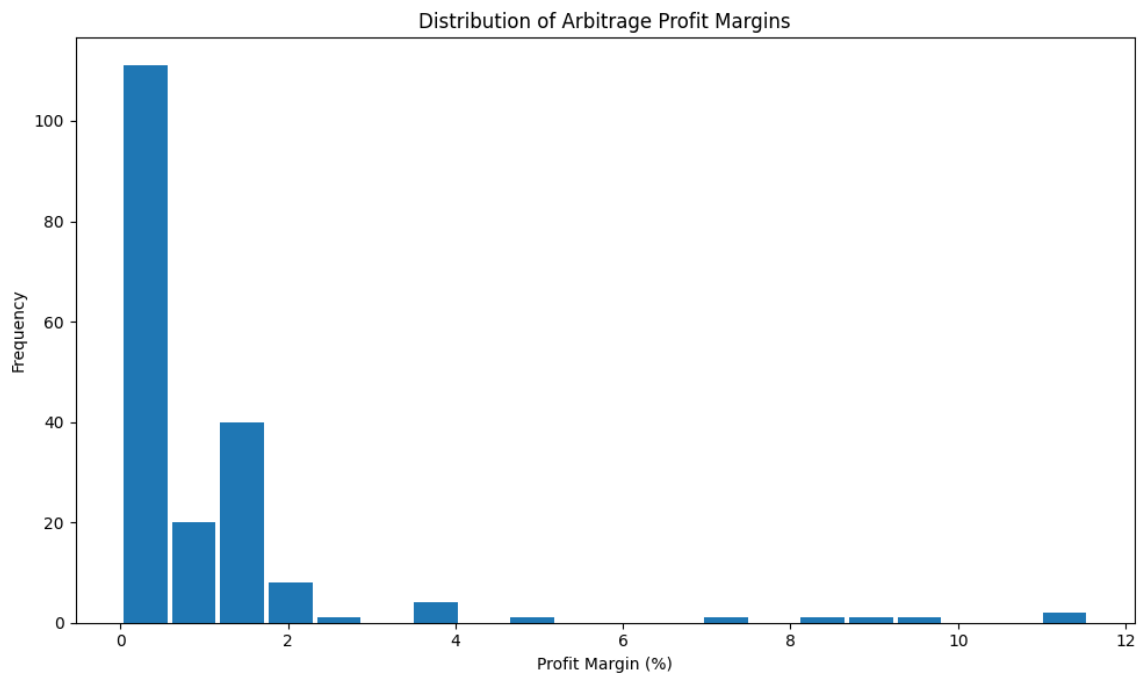


Figure 4: Distribution of Arbitrage Profit Margins

Most of the detected opportunities had profit margins below 2%, which means they would only be useful if acted on quickly and with enough stake size. However, the outliers above 10% show that larger pricing gaps did occasionally appear in the data.

4.4 Average Profit Margin by Sport

Figure 4 compares average profit margins by sport.

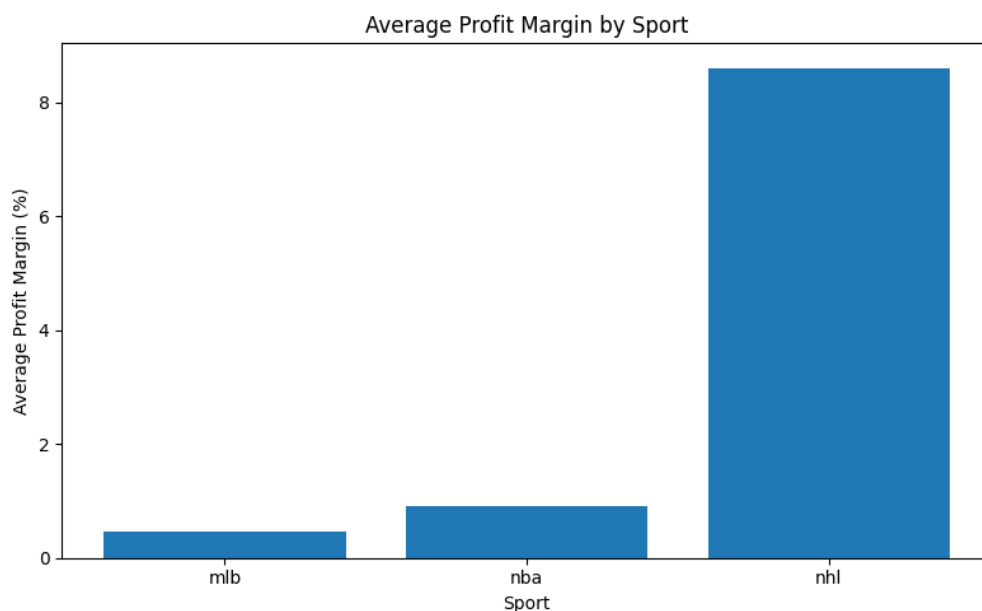


Figure 5: Average Profit Margin by Sport

Although NHL had fewer arbitrage opportunities overall, it produced the highest average profit margin. This suggests that less heavily bet sports may contain larger pricing inefficiencies.

4.5 Arbitrage Opportunities by Hour

Figure 5 shows how arbitrage opportunities varied throughout the day.

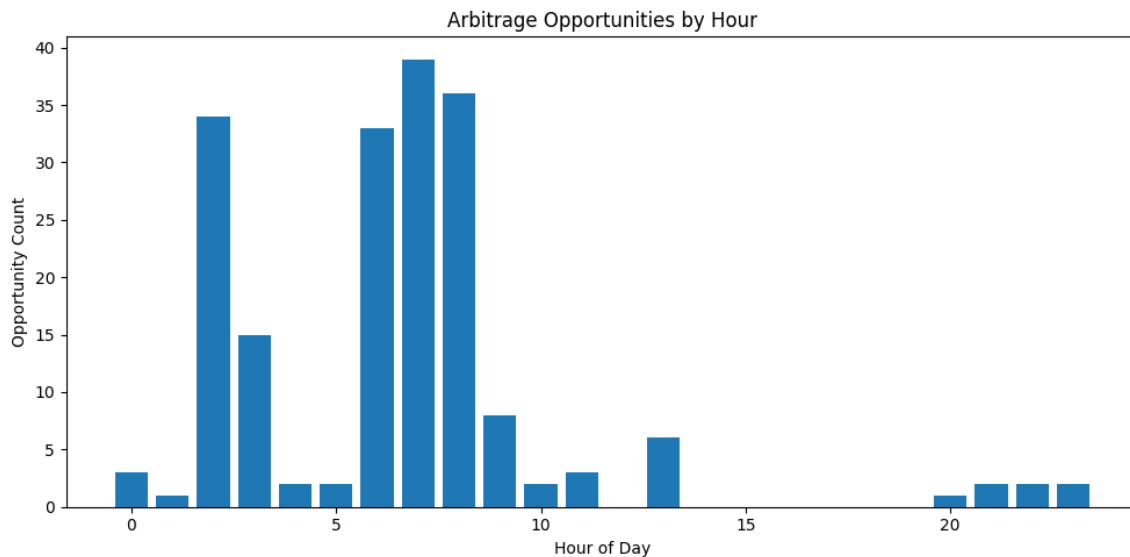


Figure 6: Arbitrage Opportunities by Hour

Most opportunities occurred during early morning hours and shortly before major sporting events began. This likely reflects periods where sportsbooks update odds asynchronously.

4.6 Top Sportsbook Pairings

Figure 6 identifies sportsbook combinations that produced the most arbitrage opportunities.

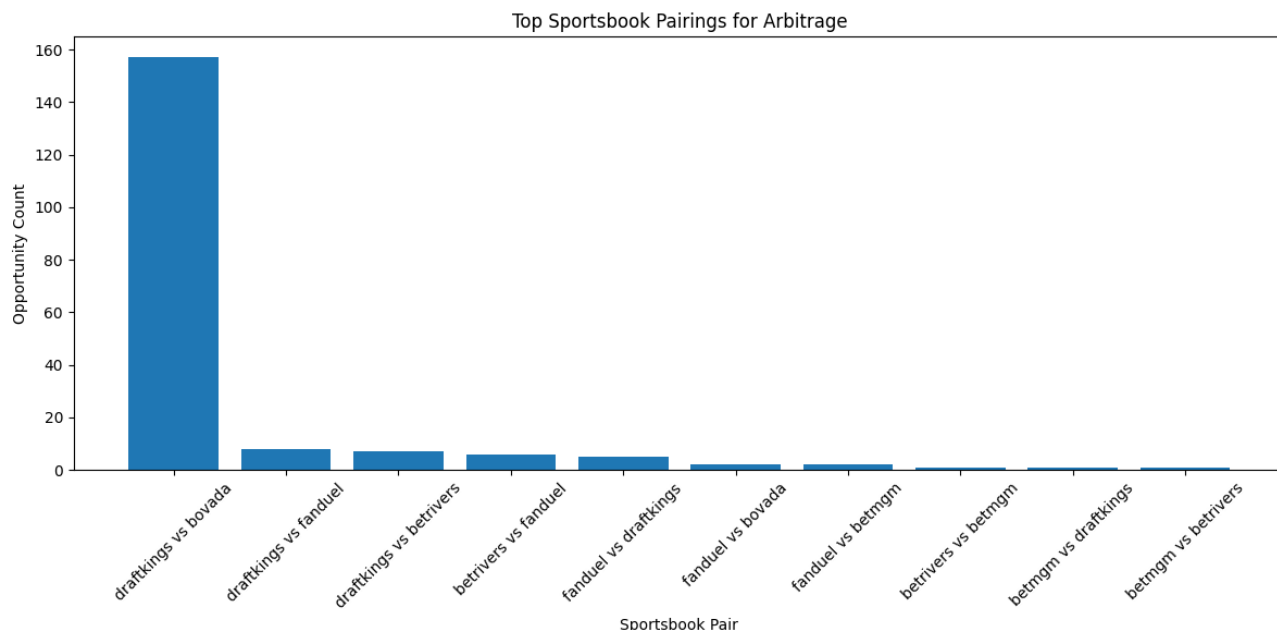


Figure 7: Top Sportsbook Pairings for Arbitrage

DraftKings and Bovada appeared together far more often than any other sportsbook pair. This does not prove that either sportsbook is consistently mispricing games, but it does show that their odds differed from each other most often in my collected dataset.

4.7 Sportsbook Pair Heatmap

Figure 7 visualizes sportsbook pair relationships using a heatmap.

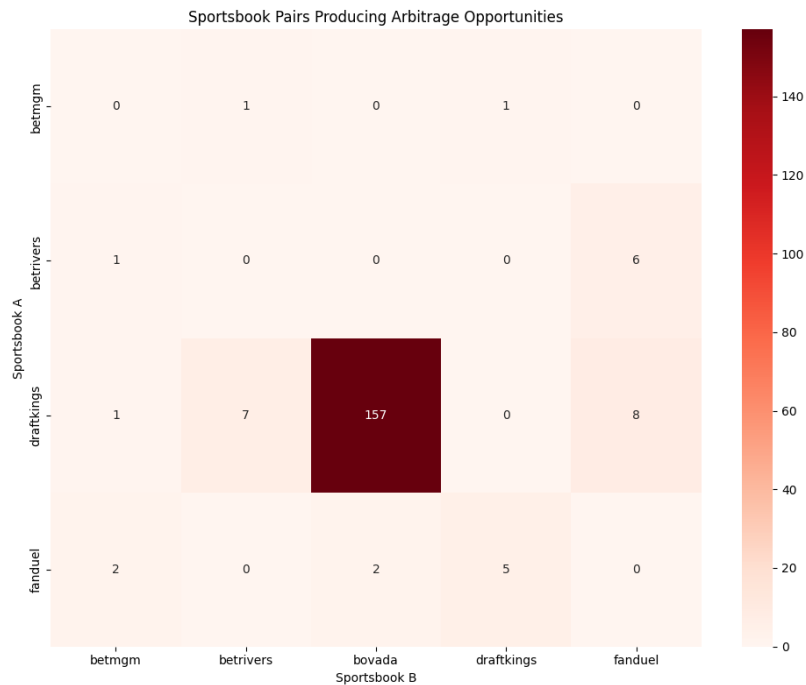


Figure 8: Sportsbook Pair Heatmap

The heatmap confirms that DraftKings and Bovada generated the highest concentration of arbitrage opportunities.

5 Conclusions

This project produced a working pipeline for collecting sportsbook odds, cleaning the data, combining records across sources, and detecting arbitrage opportunities. The final system was able to identify cases where odds from different sportsbooks created a theoretical guaranteed profit.

The results showed that arbitrage opportunities were common, but they were not evenly distributed. NBA produced the most opportunities, while NHL had the highest average profit margin in the collected data. Most profit margins were small, but a few larger outliers showed that temporary pricing gaps can be significant. DraftKings and Bovada were the most common sportsbook pairing in the detected opportunities.

The project also showed the practical difficulty of working with live web data. API data was easier to clean and maintain, while Selenium scraping required more debugging because sportsbook websites change frequently and load content dynamically. Overall, the project showed how automated data collection and integration can be used to find short-lived inefficiencies in a real market.

6 Future Work

If I continued this project, the first improvement would be making the data collection more stable. The Selenium scrapers were the weakest part of the pipeline because changes to sportsbook websites can break the selectors. A more reliable version would either add stronger error handling or rely more heavily on API-based sources.

I would also expand the project to include more sportsbooks, more sports, and more betting markets. Another useful improvement would be a real-time dashboard or alert system that notifies the user when a profitable opportunity

appears. Finally, storing the data in a relational database would make it easier to query historical odds and analyze how often arbitrage opportunities appear under different conditions.